AI Ops Module 1

Question-1

1. (a) - Entanglement (CACE — “Changing Anything Changes Everything”)
   (b) - Undeclared Consumers
   (c) - Configuration and Glue-Code Debt
2. for (c) we can use Git + DVC to make the pipeline versioned and reproducible. Convert the given undocumented shell scripts into a DVC pipeline with explicit stages, required dependencies etc. and use Git to version the pipeline configuration. This makes the workflow traceable, easier to maintain and reproducible.

Question-2

1. Screenshots for this are provided in the ”Screenshots” folder
2. The best run is mlp-hidden-256 with a learning rate of 0.001. It achieved the highest accuracy of 96.51% and an F1 score of 96.47% among the runs that I tried. This means the model correctly classified most of the MNIST test images and performed well across all digit classes. Increasing the number of hidden-layer units from 32 to 64, 128, and finally 256 generally improved the results when the learning rate was kept at 0.001. This suggests that giving the model more units helped it learn the patterns in the handwritten digits better.

   The learning rate had a much larger effect on performance than the hidden-layer size. For example, with 256 hidden units, the accuracy was 96.51% with a learning rate of 0.001, but it dropped to 66.03% with 0.01 and to only 11.45% with 0.1. In comparison, changing the hidden-layer size from 32 to 256 at a learning rate of 0.001 improved accuracy from 93.19% to 96.51%, which is a much smaller change compared to the changes occurred in learning rate
3. photo of the code is provided in Screenshots folder

Question-3

All Screenshots required are provided in ”Screenshots” folder

Question-4

Another Repository which includes reproducibility drill